

**Scenario 1: Identifying Trend**

You are given a dataset showing daily temperature for the past year. You notice the values generally increase over time. How do you confirm the presence of a trend?

1. Plot the data - Use a line plot to observe general upward/downward movement.
2. Check rolling mean - Use rolling window mean to smooth the series.
3. Use statistical test - Apply Augmented Dickey-Fuller (ADF) test:  
If  $p\text{-value} > 0.05$  - non-stationary  $\rightarrow$  likely trend exists.  
Conclusion: Data has a trend, confirmed by plot and ADF test.

**Scenario 2: Detecting Seasonality**

You are analyzing monthly electricity usage and notice similar spikes every summer. How do you confirm seasonality?

1. Plot the data over time - Use a line plot and check if spikes repeat around the same months each year (summer).
2. Use seasonal averaging / decomposition - Group by month or apply seasonal decomposition to see if the same months consistently show higher usage.
3. Confirm with a seasonal test - Check autocorrelation (ACF); strong spikes at lag 12 (for monthly data) confirm seasonality.

**Scenario 3: Making Data Stationary**

Your time series model gives poor results. The data shows trend and seasonality. How do you prepare it for ARIMA?

1. Remove trend - Apply differencing to make the series stationary.
2. Remove seasonality - Apply seasonal differencing (for monthly data, difference at lag 12).
3. Recheck stationarity - Use ADF test; if  $p\text{-value} < 0.05$ , the data is ready for ARIMA.

**Scenario 4: Choosing Model for No Seasonality**

Your time series data (e.g., daily website visits) has no visible seasonality but has a clear trend. Which model would you choose?

1. Plot the data over time - Confirm if no seasonality is visible.
2. Use ADF test  $\rightarrow$  To check stationarity.
3. Use ARIMA model (not SARIMA or Holt-Winters).
4. Use `auto_arima` or ACF/PACF to find parameters.

**Scenario 5: Evaluating Forecast Performance**

You forecast sales for 3 months and want to check how accurate it is. What steps do you follow?

1. Compare forecast with actual values - Line up predicted sales against real sales for the 3 months.
2. Calculate error metrics - Use measures like MAE, RMSE, or MAPE to quantify the forecast error.
3. Interpret the errors - Smaller error values mean better accuracy; large errors signal the model needs improvement.

### Scenario 6: Sudden Spike in Data

You're analyzing daily water consumption. One day, the value suddenly jumps very high. What will you do?

1. Verify the data point - Check if it's a data error or a real event (sensor issue, leak, special usage).
2. Analyze impact - Compare it with nearby days to see how much it affects the overall pattern.
3. Treat appropriately - If it's an error, correct or remove it; if it's real, keep it and document the reason.

### Scenario 7: Sales Goes Up Every December

You're analyzing a store's monthly sales. Sales always increase in December. What does this indicate?

1. Check if this happens every year - Yes.
2. This means there is seasonality - A pattern that repeats every year.
3. Model to use: Use models that handle seasonality like:
  - Holt-Winters
  - SARIMA
  - Facebook Prophet

Conclusion:

A consistent December sales spike shows seasonal demand, likely driven by holidays or year-end shopping.

### Scenario 8: Forecasting with Missing Values

You are forecasting stock prices, but a few days of data are missing. What should you do?

1. Identify missing dates - Check which trading days are missing and confirm they're not market holidays.
2. Handle missing values properly - Fill gaps using forward fill, interpolation, or remove those days, depending on how many are missing.
3. Validate the result - Recheck continuity and make sure the series still makes sense before forecasting.